

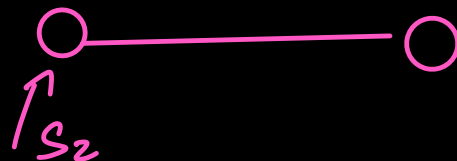
Merge Intervals

→ random order

[[s₀, e₀] [s₀, e₀] [s₀, e₀]]_n

[[4, 8], [3, 7], [1, 5], [2, 6]]





$$s_2 \leq e_1$$

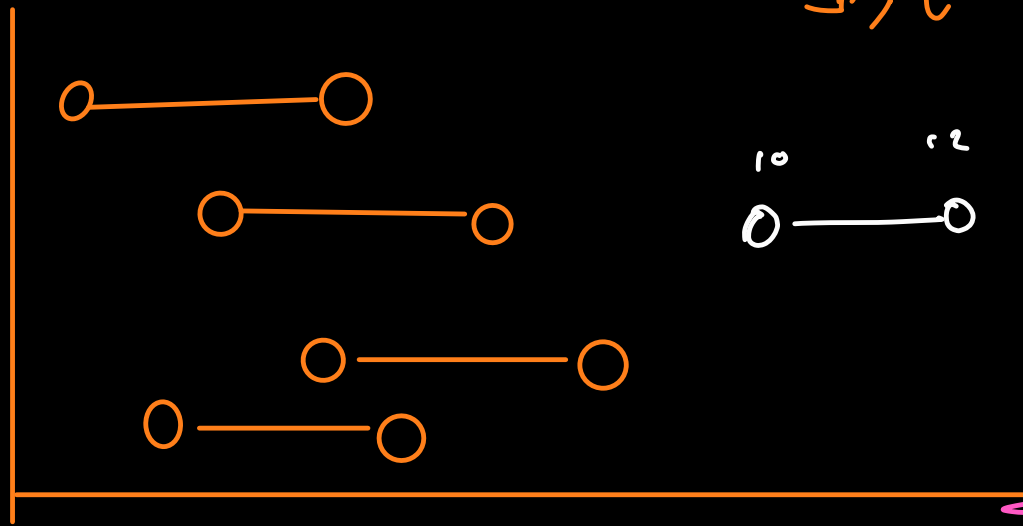
3

→ re-arrangement

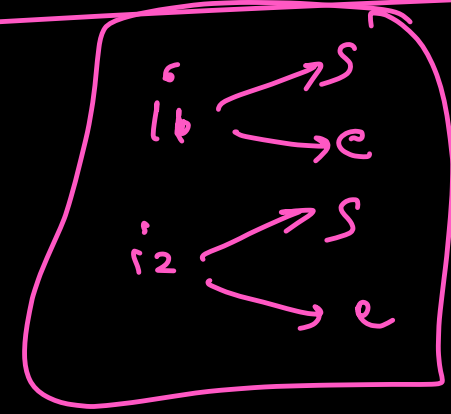
[, , ,]

[1, 5] [2, 6] [3, 7] [4, 8], [10, 12]

↓ on what parameters
we do the re-arrangement??



compare (i₁, i₂)



$[[3, 7], [1, 4], [4, 6]]$

start $[[1, 4] [3, 7] [4, 6]]$

end

$[[1, 4] [4, 6] [3, 7]]$

$[a, b]$

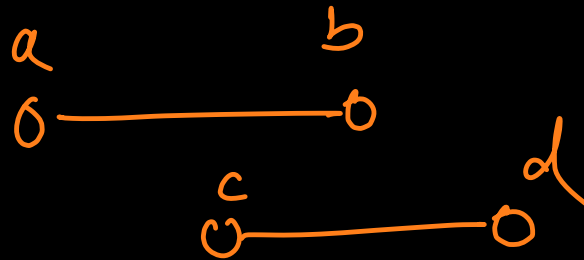
$[c, d]$

$a \leftrightarrow d$

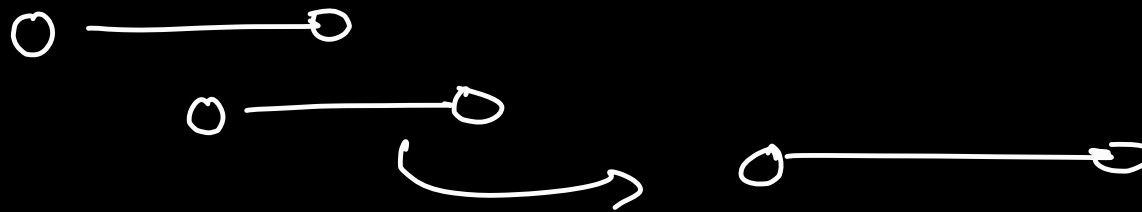
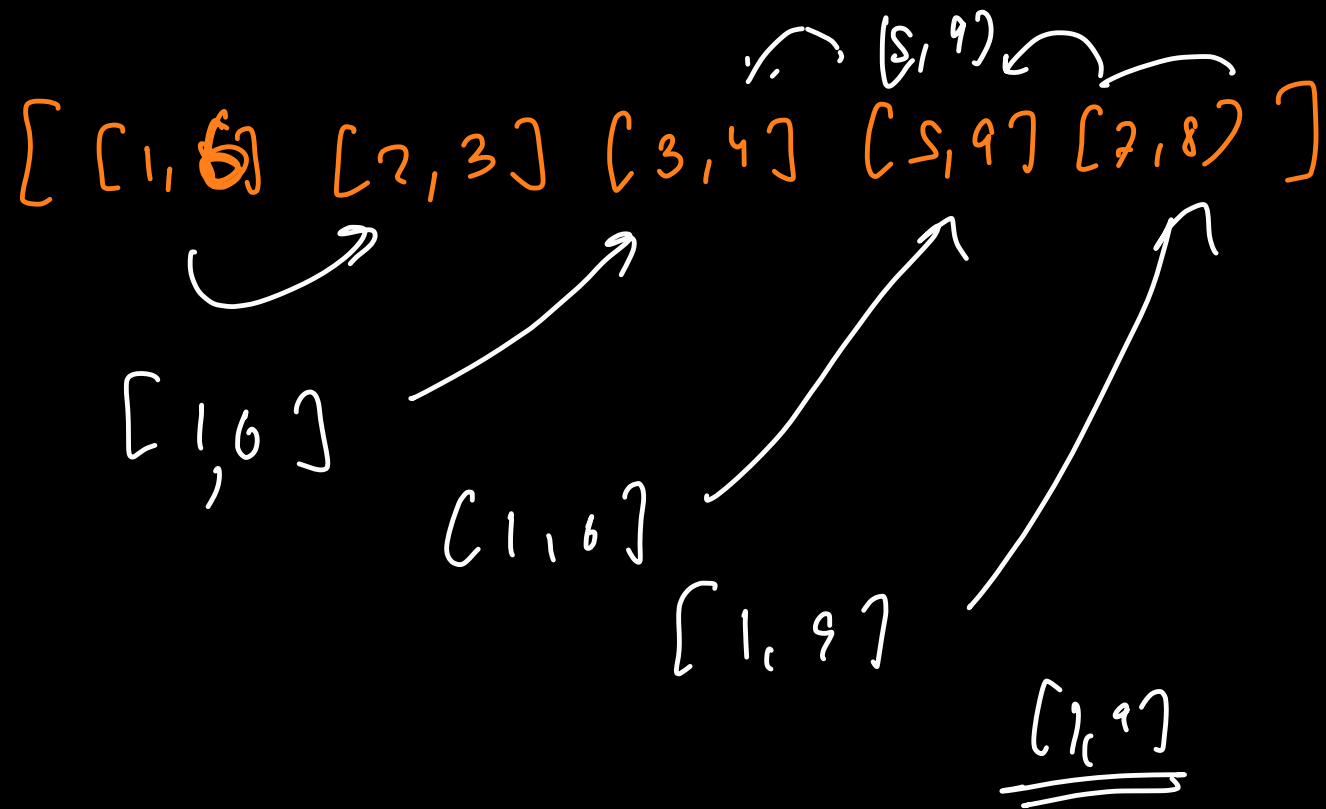
$a \leftrightarrow c$

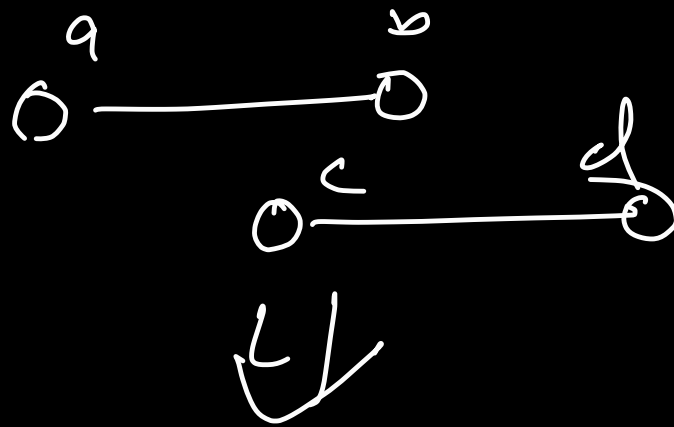
$b \leftrightarrow c$

$b \leftrightarrow d$



find $\rightarrow$ start $\rightarrow \min(a, c)$
end $\rightarrow \max(b, d)$





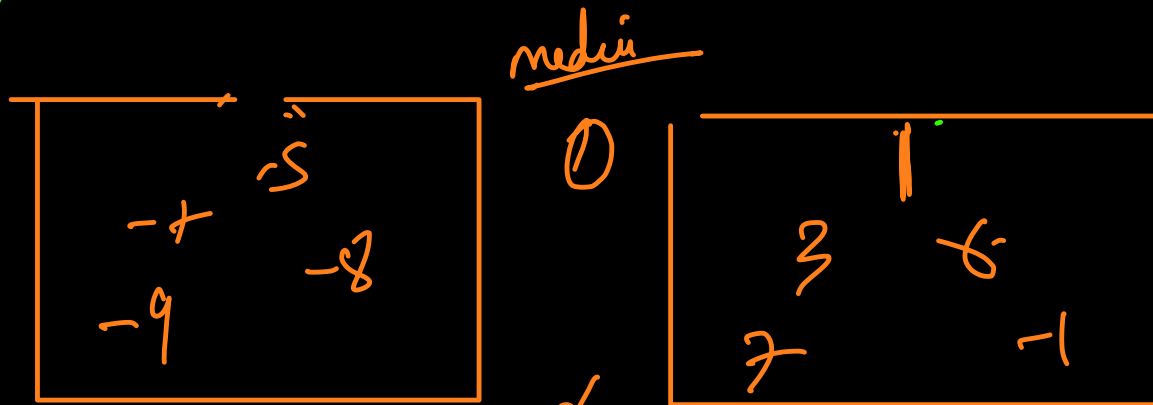
$\min(a, c)$

$\max(b, d)$



$[1, 3, -1, -3, 5, 3, 6, 7, 1, -1, -5, -7]$ $k=3$ max
 $0 \quad 1 \quad 2 \quad 3 \quad 4 \quad 5 \quad 6 \quad 7$
 $-8, -9$ $(-3, 3)$

lazy



max

$x=2$

min

$y=2, 3, 4, 5$

$$S_2 \text{ max} = S_2 \text{ min}$$

$$S_2 \text{ max fl} = S_2 \text{ min}$$

1, 3, -1, -3

-3

max

1
3

mini